INCH POUND

MIL-DTL-31011B
February 10, 2005
SUPERSEDING
MIL-DTL-31011A
December 6, 2000

DETAILED SPECIFICATION

CLOTH, WATERPROOF AND MOISTURE VAPOR PERMEABLE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for four types of camouflage cloth, which are waterproof and moisture vapor permeable.

1.2 Classification. (see 6.2)

Type I - For Bivy Covers
- Class 1 - Woodland Camouflage
- Class 2 - Desert Camouflage
- Class 4 - Universal Camouflage

Type II - For Rainsuits
- Class 1 - Woodland Camouflage
- Class 2 - Desert Camouflage
- Class 4 - Universal Camouflage

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COCT (Bldg 6), 700 Robbins Ave., Philadelphia, PA 19111-5092 or emailed to Trina.Gooding@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 8305

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Type III - For Extended Cold Weather Clothing System – Second Generation (2GECWCS) Parka and Trousers
Class 1 - Woodland Camouflage
Class 2 - Desert Camouflage
Class 4 - Universal Camouflage

Type IV - For Extended Cold Weather Clothing System (ECWCS) Parka and Trousers and for Air Force Security Police Jacket and Hood
Class 1 - Woodland Camouflage
Class 2 - Desert Camouflage
Class 3 - Air Force Blue
Class 4 - Universal Camouflage

Type V - For Extended Cold Weather Clothing System – Second Generation (Army) (2GECWCS-Army) Parka and Trousers
Class 4 - Universal Camouflage

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government Documents

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-PRF-63460 - Lubricant, Cleaner and Preservative for Weapons and Weapons Systems (METRIC)

MIL-DTL-83133 - Turbine Fuel, Aviation NATO F-34 (JP-8) and NATO F-35

STANDARDS

FEDERAL

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FED-STD-595 - Colors Used in Government Procurement

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)"

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

16 CFR Part 1500 – Federal Hazardous Substances Act Regulations
29 CFR Part 1910 – Occupational Safety and Health Standards

(Applications for copies of referenced documents should be addressed to U. S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available form the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N. W., Washington, DC 20580-0001)

CODE OF FEDERAL REGULATIONS

Title 40, part 798.4470 (Primary Dermal Irritation)

(This reference may be found on the Internet at www.access.gpo.gov/nara/cfr/cfr-table-search.html)

BUREAU OF ALCOHOL, TOBACCO AND FIREARMS, DEPARTMENT OF THE TREASURY

Formulas for Denatured Alcohol (27 CFR Part 21)

(Applications for copies of referenced document should be addressed to the Bureau of Alcohol, Tobacco and Firearms, Department of the Treasury, 1200 Pennsylvania Ave., Washington, DC 20226)

ENVIRONMENTAL PROTECTION AGENCY

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act (40 CFR Part 162)

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(Applications for copies of referenced documents should be addressed to the Environmental Protection Agency, 401 M Street, S. W., Washington, DC 20460)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER
U.S. ARMY SOLDIER SYSTEMS CENTER

2-1-1516B	-	Woodland Pattern
2-1-2240	-	3-Color Desert Pattern
2-1-2519	-	Universal Camouflage Pattern, 60-inches

(Copies of drawings are available from the U.S. Army Natick Soldier Center, Attn: AMSRD-NSC-IP-E, Natick, MA 01760-5019.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue o f the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY

ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available on line at http://www.asq.org or from the American Society for Quality Control, 600 North Plankinton Avenue, Milwaukee, WI 53203.).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC-8	- Colorfastness to Crocking: AATCC Crockmeter Method
AATCC-16	- Colorfastness to Light
AATCC-22	- Water Repellency: Spray Test
AATCC-61	- Colorfastness to Laundering, Home and Commercial: Accelerated
AATCC-96	- Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC-119	- Color Change Due to Flat Abrasion (Frosting): Screen Wire Method
AATCC-135	- Dimensional Changes in Automatic Home Laundering of Woven And Knitted Fabrics

(Applications for copies of referenced documents should be addressed to the American Association of Textile Chemists and Colorists, PO Box 12215, Research Triangle Park, NC 27709-2215)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-751 - Standard Test Method for Coated Fabrics

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ASTM D-975 - Specification for Diesel Fuel Oils
ASTM D-1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus
ASTM D-1776 - Practice for Conditioning Textiles for Testing
ASTM D-2582 - Film, Plastic and Thin Sheeting, Puncture Propagation Tear, Resistance of
ASTM D-3393 - Specification for Coated Fabrics - Waterproofness
ASTM D-3776 - Mass Per Unit Area (Weight) of Woven Fabric
ASTM D-3886 - Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Method)
ASTM D-4485 - Performance Specification for Automotive Engine Oils
ASTM D-5034 - Breaking Force and Elongation of Textile Fabrics (Grab Test)
ASTM E-96 - Test Methods for Water Vapor Transmission of Materials
ASTM F-392 - Test Method for Flex Durability of Flexible Barrier Materials

(Copies should be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI Method T-451 - Flexure Properties of Paper (Clark Stiffness)

(Applications for copies of referenced documents should be addressed to TAPPI Press, Technology Park/Atlanta, PO Box 105113, Atlanta, GA 30348-5113)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2).

3.2 Standard sample. The cloth shall match the standard sample for shade and appearance on the face side and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.3 Performance requirements. The cloth shall conform to the requirements specified in Table I and 3.4 through 3.12.

TABLE I. Physical requirements

Characteristic	Type I	II	III	IV	V
Weight, oz./sq. yd. (max)	6.0	5.0	6.0	6.0	5.8
Breaking strength, lbs. (min)					
Warp	135	135	135	135	135
Filling	100	100	100	100	100
Tearing strength, kgf (min)					

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TABLE I. Physical requirements (Continued)

Characteristic	Type I	II	III	IV	V
Warp	2.2	0.6	3.0	3.0	3.0
Filling	2.8	0.9	3.0	3.0	3.0
Hydrostatic resistance, psi (min) or No leakage (min), as applicable:					
Initial	90	90	No leakage	90	No leakage
After strength of coating	90	80	No leakage	80	No leakage
After abrasion	90	----	No leakage	80	No leakage
After high humidity	80 1/	70 1/	No leakage	----	No leakage
After diethyltoluamide					
Initial	80	70	No leakage	80	No leakage
After laundering	----	----	No leakage	----	No leakage
After diesel fuel					
Initial & after laundering	----	----	No leakage	----	----
Hydrostatic resistance, psi (min) or No leakage (min), as applicable:					
After weapons lubricant					
Initial & after laundering	----	----	No leakage	----	----
After motor oil					
Initial & after laundering	----	----	No leakage	----	----
After JP-8 fuel					
Initial & after laundering	----	----	No leakage	----	----
Moisture vapor transmission rate: $g/m^2/24$ hr (min):					
Initial					
Procedure B	400	350	600	600	600
Procedure BW	5000	5000	3600	3600	4300
After synthetic perspiration					
Procedure B	----	----	600	600	600
Procedure BW	----	----	3600	3600	4300
Stiffness, warp only, cm (max):					
At 70°F	12.0	10.0	10.0	12.0	3/
At 32°F	----	11.5	----	----	----
At 0°F	13.0	----	----	----	----
Blocking, rating (max)	No. 2	No. 2	----	----	----
Adhesion of coating 2/, lbs./2-inch width (min):					
Dry	----	12.0	----	----	----
Wet	----	5.0	----	----	----

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TABLE I. Physical requirements (Continued)

Characteristic	Type I	II	III	IV	V
Water permeability (min)					
Initial	No leakage	No leakage	No leakage	No leakage	No leakage
After synthetic perspiration					
Initial and after laundering	----	----	No leakage	No leakage	---
After physical surface appearance	No leakage	No leakage	----	----	No leakage
After flex (70°F)					
Warp and filling directions	----	----	No leakage	No leakage	No leakage
After cold flex (-25°F)					
Warp and filling directions	No leakage	----	----	No leakage	---
After cold flex (-40°F)					
Warp and filling directions	No leakage	No leakage	No leakage	No leakage	No leakage
After diethyltoluamide					
Initial and after laundering	No leakage	No leakage	No leakage	No leakage	No leakage
After diesel fuel					
Initial and after laundering	No leakage	No leakage	No leakage	No leakage	---
Water permeability (min)					
After weapons lubricant					
Initial and after laundering	No leakage	No leakage	No leakage	No leakage	----
After motor oil					
Initial and after laundering	No leakage	No leakage	No leakage	No leakage	----
After JP-8 fuel					
Initial and after laundering	No leakage	No leakage	No leakage	No leakage	----

1/ The cloth shall not become stiff or brittle nor soft and tacky and there shall be no evidence of cracking or crazing under visual examination.
2/ Applicable to fabrics with a continuous film or coating.
3/ The report shall include only the result for each of the test specimens; no individual test specimen result shall be greater than 7.5 centimeters.

3.4 Spray rating. Equal to or better than 100,90, 90 initially (Types I, II, IV and V) and 90, 90, 80 after 3 launderings (Types I, II, III, and IV) and 90, 90, 90 after 5 launderings (Type V). Not applicable to fabrics with a continuous film or coating on the face side.

3.5 Resistance to organic liquids (All Types). The cloth shall show no wetting initially and after laundering. Not applicable to fabrics with a continuous film or coating on the face side.

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3.6 Physical surface appearance after laundering. The cloth shall show no change in physical surface appearance after 5 laundering and drying cycles (Types I and II) or 20 laundering and drying cycles (Types III, IV and V).

3.7 Dimensional stability (Types III, IV and V only). The shrinkage or elongation of the cloth shall not be greater than 4.0 percent in the warp direction and not greater than 2.0 percent in the filling direction for Types III and IV, and not be greater than 4.0 percent in each of the warp and filling directions for Type V.

3.8 Colorfastness. The face side of the cloth shall meet the following colorfastness requirements:

Colorfastness characteristics	Requirements
Fastness to laundering	Equal to or better than "3-4" rating on AATCC Gray Scale for Color Change
Fastness to accelerated laundering (Black print only)	Equal to or better than "3-4" rating on AATCC Gray Scale for Color Change
Fastness to light	Equal to or better than "3-4" rating on AATCC Gray Scale for Color Change
Fastness to crocking	Equal to or better than the standard sample or not less than AATCC chromatic transference scale rating of 3.5. 1/
Fastness to abrasion	Equal to or better than the standard sample or not less than "3-4" rating on AATCC Gray Scale for Color Change
Fastness to high humidity (except Types IV and V)	No appreciable change 2/

1/ Except Black 357 shall show an AATCC chromatic transference scale rating of not less than 1.0.
2/ An appreciable change in color means a change that is immediately noticeable on comparison of the test specimen with the original, unexposed sample. If closer inspection or a change of angle of light is required to make apparent a slight change of color, then change is not considered appreciable.

3.9 Color

3.9.1 Class 1, Woodland Camouflage. The color of the face side of the cloth shall be Woodland Camouflage pattern and shall match Light Green 354, Dark Green 355, Brown 356, and Black 357. Each area of the specific color of the pattern shall be in accordance with the applicable standard sample. The color of the back side of the cloth shall be Camouflage Green 483 matching color chip 34094 of FED-STD-595B or show Woodland Camouflage strike-thru from the face side print.. Both sides of the cloth shall match the respective side of the standard sample.

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3.9.2 Class 2, Desert Camouflage. The color of the face side of the cloth shall be Desert Camouflage pattern and shall match Light Tan 492, Light Khaki 494, and light Brown 493. Each area of the specific color of the pattern shall be in accordance with the applicable standard sample. The color of the back side of the cloth shall be Light Tan 492 or show Desert Camouflage strike-thru from the face side print.. Both sides of the cloth shall match the respective side of the standard sample.

3.9.3 Class 3, Air Force Blue. The color of the face side of the cloth shall be Air Force Blue, Shade 1613 (see 6.2). The color of the back side of the cloth shall approximate Air Force Blue Shade 1613.

3.9.4 Class 4, Universal Camouflage. The color of the face side of the cloth shall be Universal Camouflage pattern and shall match Desert Sand 500, Urban Gray 501, and Foliage Green 502. Each area of the specific color of the pattern shall be in accordance with the applicable standard sample. The color of the back side of the cloth shall be Foliage Green 504 or show Universal Camouflage strike-thru from the face side print. Both sides of the cloth shall match the respective side of the standard sample.

3.10 Pattern Execution.

3.10.1 Class 1, Woodland Camouflage pattern execution. The Woodland Camouflage pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The pattern repeat of the dyed, printed, and finished cloth shall be 27.25 inches (+2.50 inches, -2.50 inches) in the warp direction. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas shall not be acceptable. When the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided, and the pattern on the finished cloth shall match that of drawing 2-1-1516B.

3.10.1 Class 2, Desert Camouflage pattern execution. The Desert Camouflage pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The pattern repeat of the dyed, printed, and finished cloth shall be 16.75 inches (+1.25 inches, -1.75 inches) in the warp direction. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas shall not be acceptable. When the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided, and the pattern on the finished cloth shall match that of drawing 2-1-2240.

3.10.1 Class 4, Universal Camouflage pattern execution. The Universal Camouflage pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The pattern repeat of the dyed, printed, and finished cloth shall be 36.00 inches (+1.25 inches, -2.50 inches) in the warp direction. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that

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shown by the standard sample in any of the printed areas shall not be acceptable. When the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided, and the pattern on the finished cloth shall match that of Drawing 2-1-2519.

3.11 Spectral Reflectance.

3.11.1 Spectral reflectance, Class 1 Woodland Camouflage. The spectral reflectance of the colors in the woodland camouflage cloth shall conform to the requirements specified in Table II, initially (all types) and after laundering (Types II, III, and IV) when tested as specified in 4.4.16.

3.11.2 Spectral reflectance, Class 2 Desert Camouflage. The spectral reflectance of the colors in the desert camouflage cloth shall conform to the requirements specified in Table III, initially (all types) and after laundering (Types II, III, and IV) when tested as specified in 4.4.16.

3.11.3 Spectral reflectance, Class 4 Universal Camouflage. The spectral reflectance of the colors in the Universal Camouflage cloth shall conform to the requirements specified in Table IV, initially and after laundering when tested as specified in 4.4.16.

TABLE II. Spectral reflectance requirements; Reflectance (percent)

Wavelengths	Black 357		Light Green 354		Dark Green 355 & Brown 356	
Nanometers (nm)	Min	Max	Min	Max	Min	Max
600	-	-	8	20	3	9
620	-	-	8	20	3	9
640	-	-	8	20	3	9
660	-	-	8	20	3	12
680	-	-	10	30	3	16
700	-	20	18	50	5	32
720	-	30	22	54	7	44
740	-	33	30	56	12	52
760	-	33	35	58	18	56
780	-	34	40	62	26	56
800	-	34	55	80	34	56
820	-	35	55	80	42	60
840	-	35	55	84	44	60
860	-	35	60	84	44	60

TABLE III. Spectral reflectance requirements; Reflectance (percent)

Wavelengths	Light Tan 492		Light Brown 493		Light Khaki 494	
Nanometers (nm)	Min	Max	Min	Max	Min	Max
700	38	53	19	36	25	48
720	38	58	20	36	25	52
740	39	62	20	36	25	54
760	40	66	21	36	26	56
780	41	72	21	38	27	57
800	43	76	22	43	28	58

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820	45	76	23	45	30	58
840	48	78	24	46	33	58
860	50	78	25	46	36	59

TABLE IV. Spectral reflectance requirements; Reflectance (percent)

Wavelengths	Desert Sand 500		Urban Gray 501		Foliage Green 502	
Nanometers (nm)	Min	Max	Min	Max	Min	Max
600	28	42	12	26	8	18
620	30	44	14	26	8	18
640	34	50	14	28	8	20
660	38	59	14	30	10	26
680	44	63	18	34	10	26
700	46	69	24	38	12	28
720	48	71	26	42	16	30
740	48	76	30	46	16	30
760	50	80	32	48	18	32
780	54	80	34	48	18	34
800	54	80	34	50	20	36
820	54	80	36	54	22	38
840	56	82	38	54	24	40
860	56	82	40	56	26	42

3.12 Toxicity. The finished cloth shall not present a dermal health hazard when used as intended (see 4.4.18).

3.13 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.2)
2. Conformance inspection (see 4.3)

4.2 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected as specified in 4.3.2 through 4.4 for compliance with design, construction, workmanship and dimensional requirements.

4.3 Conformance inspection. Sampling for inspection shall be performed in accordance with ASQC Z1.4 and with quality acceptance levels as specified in the contract and/or order, except where otherwise indicated.

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4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1. The cloth shall be examined for the defects in Table V. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The number of rolls from which the sample yardage is to be selected shall be in accordance with the table below. The sample yardage shall be apportioned equally among the selected rolls.

Lot size in yards	Sample size in rolls
1,200 or less 1/	3
1,201 up to and including 3,200	5
3,201 up to and including 10,000	8
10,001 up to and including 35,000	13
35,001 up to and including 150,000	20
150,001 and over	32

1/ If lot contains fewer than three rolls, each roll in the lot shall be examined.

4.3.2.1 Visual examination. The cloth shall be examined (on both sides) for the defects listed in Table V.

Table V End item visual defects

Defect	Classification	
	Major	Minor
Any hole, cut tear or scratch, including edges	101	
Abrasion resulting in a thin or weak place	102	
Multiple floats or skips, ½ inch or more in either warp or filling direction of face fabric	103	
Any pit, blister, tunnel, bubbles, or delamination of components	104	
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure, adhesion of surfaces against each other, or any diagonal distortion of face side surface	105	
Any solid lump, defined as a slub C or 4, or knot which exceeds level C on the respective Sears Fabric Defect Scale (see 2.3)	106	
Fabric edges rolled, folded, doubled, scalloped or wavy	107	
Any spot, stain 1/ or foreign matter 2/	108	

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Any odor other than that which is characteristic of the component materials of the cloth		201
Any color off shade, not uniform, mottled or spotted (face side only)	109	
Any tackiness	110	

Table V End item visual defects (Cont'd)

Defect	Classification Major	Minor
Any pinhole	111	
Any area without waterproof finish (i.e., coating or laminating film, where required)	112	
Any scorch or burn	113	
Multiple floates or skips, ½ inch or more in either warp or filling direction of face fabric	114	
Not clean	115	
Camouflage pattern (face side)		
Any skitteriness of pattern exceeding that shown by the standard sample	116	
Pattern design not equal to standard sample	117	
Excessive feathering or spew of pattern	118	
Pattern repeat not equal to the standard sample	119	
Woodland Camouflage pattern less than 24.75 inches or more than 29.75 inches	120	
Desert Camouflage pattern less than 15.00 inches or more than 18.00 inches	121	
Universal Camouflage pattern less than 33-1/2 inches or more than 37-1/4 inches	122	

1/ Clearly visible at normal inspection distance (approx. 3 ft.).
2/ For the back side, any spot, stain, off-shade area, or discoloration that is a result of the distortion of a backing fabric (if used) or a result of uneven dyeing of a backing fabric shall not be scored for this condition. Foreign matter shall be defined as waste, fly or extraneous material that has been formed into the fabric system.

4.3.3 End item testing. The cloth shall be tested for the characteristics listed in Table VI. The methods of testing specified wherever applicable and as listed in Table VI shall be followed. The sample unit for testing shall be 6 continuous yards full width of the finished cloth, put up in a manner to prevent folding/creasing. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be in accordance wit the following:

TABLE VI. End item testing

Characteristic	Requirement paragraph	Test methods*
Weight	3.3	ASTM D-3776 (Method C)

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Breaking strength	3.3	ASTM D-5034 (G-E or G-T)
Tearing strength		
Types I, III, IV and V	3.3	4.4.1.1
Type II	3.3	4.4.1.2
Hydrostatic resistance		
Initial		
Types I, II and IV	3.3	4.4.2.1
Type III and V	3.3	4.4.2.2
After strength of coating	3.3	4.4.2.3
After abrasion	3.3	4.4.2.4
After high humidity	3.3	4.4.2.5
After diethyltoluamide		
Initial		
Types I, II and IV	3.3	4.4.2.6
Type III and V	3.3	4.4.13
After laundering		
Types I, II and IV	3.3	4.4.13, 4.4.14 & 4.4.2.1
Type III and V	3.3	4.4.13, 4.4.14 & 4.4.2.2
After diesel fuel		
Initial & After laundering	3.3	4.4.13, 4.4.14 & 4.4.2.2
After weapons lubricant		
Initial & After laundering	3.3	4.4.13, 4.4.14 & 4.4.2.2
After motor oil		
Initial & After laundering	3.3	4.4.13, 4.4.14 & 4.4.2.2
After JP-8 fuel		
Initial & After laundering	3.3	4.4.13, 4.4.14 & 4.4.2.2
Moisture vapor transmission rate:		
Initial		
Procedure B	3.3	4.4.8.1
Procedure BW	3.3	4.4.8.2
After synthetic perspiration		
Procedure B	3.3	4.4.12 & 4.4.8.1
Procedure BW	3.3	4.4.12 & 4.4.8.2

TABLE VI. End item testing (Cont'd)

Characteristic	Requirement paragraph	Test methods*
Stiffness:		

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At 70°F	3.3	4.4.3.1
At 32°F	3.3	4.4.3.2
At 0°F	3.3	4.4.3.3
Blocking	3.3	4.4.5
Adhesion of coating:		
Dry	3.3	4.4.6.1
Wet	3.3	4.4.6.2
Water permability:		
Initial	3.3	4.4.7
After synthetic perspiration	3.3	4.4.12 & 4.4.7
After physical surface appearance	3.3	4.4.4 & 4.4.7
After flex (70°F)	3.3	4.4.7.1 & 4.4.7
After cold flex (-25°F)	3.3	4.4.7.2 & 4.4.7
After cold flex (-40°F)	3.3	4.4.7.2 & 4.4.7
After diethyltoluamide		
Initial and After laundering	3.3	4.4.13, 4.4.14 & 4.4.7
After diesel fuel		
Initial and After laundering	3.3	4.4.13, 4.4.14 & 4.4.7
After weapons lubricant		
Initial and After laundering	3.3	4.4.13, 4.4.14 & 4.4.7
After motor oil		
Initial and After laundering	3.3	4.4.13, 4.4.14 & 4.4.7
After JP-8 fuel		
Initial and After laundering	3.3	4.4.13, 4.4.14 & 4.4.7
Spray rating		
Initial	3.4	4.4.9.1
After 3 launderings	3.4	4.4.9.2 & 4.4.9.1
After 5 launderings	3.4	4.4.9.3 & 4.4.9.1
Resistance to organic liquids		
Initial	3.5	4.4.10.1
After 3 launderings	3.5	4.4.10.2 & 4.4.10.1
After 5 launderings	3.5	4.4.10.3 & 4.4.10.1
Physical surface appearance	3.6	4.4.4
Dimensional stability	3.7	AATCC-96, Option 1C
Colorfastness to:		
Laundering	3.8	4.4.11.1
Accelerated laundering (Black only)	3.8	4.4.11.2
Light	3.8	4.4.11.3
Crocking	3.8	AATCC-8
Abrasion (Black only)	3.8	4.4.11.4
High humidity	3.8	4.4.2.5
Color matching	3.9	4.4.17

TABLE VI. End item testing (Cont'd)

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Characteristic	Requirement paragraph	Test methods*
Pattern execution	3.10	4.4.15
Spectral reflectance	3.11	4.4.16
Spectral reflectance after accelerated laundering	3.11	4.4.16.1
Toxicity	3.12	4.4.18

* Paragraph number alone indicates the paragraph number in section 4 of this document.

4.4 Methods of inspection.

4.4.1 Tear strength.

4.4.1.1 Types I, III, IV & V. ASTM D-2582, with exceptions as follows: Five warp and five filling specimens shall be tested. Specimen size shall be 8-inches by 8-inches. Only one tear shall be made on a single specimen. The specimen shall be positioned with the face side toward the probe and with the designated yarns of the face fabric at right angles to the direction of tear. The test shall be conducted using the standard drop height of 508 $\pm$ 2 mm. If the tear is not straight on face side of the specimen, the result shall be considered invalid and another specimen shall be tested. The thickness of the specimen shall not be measured.

4.4.1.2 Type II. ASTM D-1424.

4.4.2 Hydrostatic resistance.

4.4.2.1 Types I, II & IV. ASTM D-751, Hydrostatic Resistance, Procedure A (Pressure Application by Mullen Type Hydrostatic Tester), Procedure 1 with water pressure applied to the face side of the test specimen.

4.4.2.2 Type III and V. ASTM D-3393, except that a water pressure of 40 psi shall be used and applied to the face side of the test specimen.

4.4.2.3 Hydrostatic resistance after strength of coating. ASTM D-751, except that the testing machine clamp separation rate shall be 5 mm/sec (12 in/min). Hydrostatic resistance shall be determined in accordance with 4.4.2.1 for Type I, II and IV cloths and 4.4.2.2 for Type III and V cloths.

4.4.2.4 Hydrostatic resistance after abrasion. ASTM D-3886, except that the test shall be conducted in the multidirectional mode as described below. The specimens shall then be tested for hydrostatic resistance in accordance with 4.4.2.1 for Type I and IV cloth and 4.4.2.2 for Type III and V cloths.

a. Type I. The back side of the cloth shall be abraded for 1,000 cycles using the back side of the test material as the abradant.

b. Types III, IV & V. The face side of the cloth shall be abraded for 10,000 cycles using the face side of the test material as the abradant and a load of six (6) pounds.

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4.4.2.5 Hydrostatic resistance after high humidity. Three (3), 4-inch by 4-inch specimens shall be laid flat, face side up, on a supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1-inch below the specimens. The lid of the desiccator shall be put in place and the desiccator placed in a circulating air oven having a temperature of 125 ± 2°F for a period of 7 days. At the end of the aging period, each specimen shall be removed from the desiccator and then immediately examined for colorfastness and tested for hydrostatic resistance. The specimens shall be tested for hydrostatic resistance in accordance with 4.4.2.1 for Type I and II cloths and 4.4.2.2 for Type III and V cloth.

4.4.2.6 Hydrostatic resistance after diethyltoluamide (Types I, II and IV). Five (5) specimens shall be laid flat, face side up on a 4-inch by 4-inch by ¼-inch, glass plate. Three (3) drops of diethyltoluamide containing 75% diethyltoluamide and 25% ethanol (see 6.6) shall be applied to the center of each specimen. A glass plate of the same dimensions shall be placed on the specimen (or specimen assembly). After 16 hours, the specimens shall be removed from between the glass plates and tested immediately for hydrostatic resistance in accordance with 4.4.2.1.

4.4.3 Stiffness.

4.4.3.1 Stiffness at 70°F. TAPPI Method T-451, Preferred Procedure (1), except that five (5) test specimens with the long dimension parallel to the warp direction of the cloth shall be tested and that the standard textile test conditions as specified in ASTM D-1776 shall be used.

4.4.3.2 Stiffness at 32°F (Type II only). The stiffness test shall be conducted as specified in 4.4.3.1 except that the apparatus and test specimens shall be subjected to a temperature of 32° ± 2°F for a period of 4 hours and the test shall be performed in a still atmosphere at that temperature.

4.4.3.3 Stiffness at 0°F (Type I only). The stiffness test shall be conducted as specified in 4.4.3.1 except that the apparatus and test specimens shall be subjected to a temperature of 0° ± 2°F for a period of 4 hours and the test shall be performed in a still atmosphere at that temperature.

4.4.4 Physical surface appearance. Conduct five (5) laundering and drying cycles (Types I & II) or twenty (20) laundering and drying cycles (Types III, IV & V) in accordance with 4.4.14. Each sample, 48-inches in length by full width, shall be cut in half across the width of the cloth. One half of the sample (24-inches in length) shall be laundered and the remaining half retained as the unlaundered portion for the final evaluation, as necessary. After each drying cycle, examine both sides of the cloth for changes in physical surface appearance when compared to the unlaundered sample.

4.4.5 Blocking. ASTM D-751, Blocking Resistance at Elevated Temperatures, except that the tests shall be performed at a temperature of 180° ± 2°F for 30 minutes. Only one (1) specimen shall be tested. Evaluate the resistance of the specimen to blocking by the scale given below:

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1 -- *No Blocking*. Cloth surfaces are free and separate without any evidence of cohesion or adhesion.
2 -- *Trace Blocking*. Cloth surfaces show slight cohesion or adhesion.
3. -- *Slight Blocking*. Cloth surfaces must be lightly peeled to separate.
4 -- *Blocking*. Cloth surfaces separate with difficulty or coating is removed during separation.

4.4.6 Adhesion of coating.

4.4.6.1 Dry adhesion. ASTM D-751, Adhesion of Coating, except with 2-inch wide, reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/s. Three (3) specimens shall be tested by adhering the film/coating sides of the cloth to itself.

4.4.6.2 Wet adhesion. The test specimens of 4.4.6.1, after dry adhesion testing, shall be immersed in distilled water at 70°F for 16 hours, removed from the water and blotted dry and then immediately tested for wet adhesion in accordance with 4.4.6.1 using the remainder of the specimen.

4.4.7 Water permeability. ASTM D-751, Hydrostatic Resistance, Procedure B, Procedure 2 with a fixed hydrostatic head of 50 centimeters applied to the face side of the test specimen for 10 minutes. Five (5) specimens shall be tested. The report shall only include measurement for the appearance of water droplets. For Types I, III, IV and V cloths, leakage is defined as the appearance of one (1) or more droplets of water within the 4-1/2 inch diameter test area. For Type II cloth, leakage is defined as the appearance of water at three (3) or more different places or the continuous flow of water through the cloth at one (1) or more different places within the 4-1/2 inch diameter test area.

4.4.7.1 Water permeability after flex at 70°F. One specimen, 8-inches by 12-inches, shall be cut from the sample unit with the 8-inch dimension in the indicated direction (warp or filling, as applicable). The specimen shall be conditioned and flexed as specified in ASTM F-392, except that the specimen shall not be aged, the short edges shall not be heat sealed or otherwise joined, and the specimen shall be flexed for 1500 cycles. Two (2), 6-inch by 8-inch specimens shall be cut from the 8-inch by 12-inch flexed specimen and tested for water permeability in accordance with 4.4.7.

4.4.7.2 Water permeability after cold flex at -25°F and –40°F. One specimen, 8-inches by 12-inches, shall be cut from the sample unit with the 8-inch dimension in the indicated direction (warp or filling, as applicable). The specimen shall be flexed as specified in ASTM F-392, except that the specimen shall not be aged, the short edges shall not be heat sealed or otherwise joined, and the specimen shall be flexed for 1500 cycles. Prior to flexing, the specimen shall be mounted on the flex test apparatus, placed in at test chamber at the specified temperature (± 5°F) for one (1) hour, and then flexed in the test chamber at the specified temperature. At the end of the flexing cycle, two (2), 6-inch by 8-inch specimens shall be cut from the 8-inch by 12-inch flexed specimen, conditioned prior to testing, and tested for water permeability in accordance with 4.4.7.

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4.4.8 Moisture Vapor Transmission Rate. ASTM E-96 with temperature and humidity conditions of $73.5^{o} \pm 1^{o}$ F and 50 ± 2% R.H. The linear air flow velocity in the wind tunnel shall be set to yield an upright, 'open cup' evaporation rate at all test specimen positions of 15000 ± 1000 $g/m^2/24h$ (The evaporation rate shall be determined by conducting an upright cup, Procedure B test (see 4.4.8.1), without a test specimen for a period of exactly two (2) hours). For specimen testing using Procedure BW, the 'open cup' evaporation rate shall be determined in the air stream at a level not more than three (3) inches below the position of the inverted cup test specimen.

4.4.8.1 Procedure B. ASTM E-96, Procedure B. The back side of the basic material shall face the water. The test specimen shall be conditioned, after set-up in the test cup with water level of 3/4 ± 1/16 inch below the specimen surface, in the wind tunnel for a period of not less than four (4) hours and not more than sixteen (16) hours (Conditioning time of less than 4 hours may be used provided that equilibrium conditions have be demonstrated to exist within the test sample/sample cup/wind tunnel. In cases of dispute, the conditioning time shall be 4 hours.) After conditioning, the cup shall be immediately weighed to start the test and again after exactly twenty-four (24) hours to complete the test. Five (5) specimens shall be tested.

4.4.8.2 Procedure BW. ASTM E-96, Procedure BW. The back side of the basic material shall face the water. The test specimen shall be set-up in the test cup with water level of 3/4 ± 1/16 inch below the specimen surface. The cup shall be inverted bringing the water in contact with the back side of the test specimen; the test specimen shall be conditioned in this manner for five (5) minutes. At the end of the conditioning period, the cup and test specimen shall be examined for evidence of water leakage at cup edges or through holes/pinholes in the test specimen; if a leak occurs, the test specimen shall be discarded and the test cup shall be set up with a new test specimen. The cup shall be immediately weighed and placed in the wind tunnel to start the test and removed and weighed again after exactly two (2) hours to complete the test. Five (5) specimens shall be tested.

4.4.9 Spray rating.

4.4.9.1 Initial. Testing shall be conducted in accordance with AATCC-22.

4.4.9.2 After 3 launderings. Test specimens shall be laundered for three (3) laundering cycles in accordance with 4.4.14 and then tested for spray rating in accordance with 4.4.9.1.

4.4.9.3 After 5 launderings. Test specimens shall be laundered for five (5) laundering cycles in accordance with 4.4.14 and then tested for spray rating in accordance with 4.4.9.1.

4.4.10 Resistance to organic liquids.
4.4.10.1 Initial. Place a small specimen of the cloth on a smooth horizontal surface, face side up. Using a pipette or eyedropper, gently deposit one (1) drop of n-tetradecane on the surface of the specimen. After 30 seconds, examine the specimen under light at an angle. Absence of light reflectance at the cloth/drop interface shall be taken as evidence of wetting. Three (3) specimens

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(or areas) taken at various locations across the sample unit shall be tested. Evidence of wetting on one (1) or more specimens shall be considered a test failure.

4.4.10.2 After 3 launderings (Types I, II, III, and IV). Test specimens shall be laundered for three (3) laundering cycles in accordance with 4.4.14 and then tested for resistance to organic liquids in accordance with 4.4.10.1.

4.4.10.3 After 5 launderings (Type V). Test specimens shall be laundered for five (5) laundering cycles in accordance with 4.4.14 and then tested for resistance to organic liquids in accordance with 4.4.10.1.

4.4.11 Colorfastness.

4.4.11.1 Laundering. AATCC-61, Test 1A (3 cycles) except that 1993 AATCC Standard Reference Detergent (non-phosphate) without optical brighteners shall be used.

4.4.11.2 Accelerated laundering (Black 357 only). The test procedure shall be as follows using the test equipment cited in AATCC-61.

Five (5) specimens containing predominantly Black print, each 4-1/2 inches by 3 inches, shall be cut from the test fabric and then folded in half, with the face side out, to form a bag 2-1/4 inches by 3 inches. Machine stitch the open edges together (seam allowance no more than 1/4 –inch) to form a bag leaving an opening approximately 1-inch in length. Through the opening, add thirty-five (35) stainless steel spheres.

Close the bag by stitching. Place the bag in a stainless steel cylinder (one (1) bag per cylinder) without the color transfer cloth, add 50 ml of 1993 AATCC Standard Reference Detergent (non-phosphate) without optical brighteners (0.5 percent by weight) and 100 stainless steel spheres and close tightly. Place the stainless steel cylinder in a preheated Launder-Ometer set at a water bath temperature of $160^{o} \pm 5^{o}$F. Agitate cylinder for one (1) hour, maintaining a constant temperature. At the end of the laundering cycle, remove the bag from cylinder and rinse each bag thoroughly in a beaker, in running tap water at $100^{o} \pm 5^{o}$F for five (5) minutes with occasional stirring or hand squeezing. Remove excess water by squeezing in hand (not extracting) and then dry bag in automatic tumble dryer set on permanent press cycle (150°-160°F) for fifteen (15) minutes (multiple bags may be dried together). If the bag breaks open to release the contained spheres at any time during the test, the test shall be considered invalid and another bag specimen shall be prepared and tested. Remove all spheres from the bag and evaluate each face of the bag without pressing or ironing the bag. Each face of the laundered bag shall be compared to the original unlaundered sample in accordance with AATCC Evaluation Procedure 1 for evaluation of Gray Scale for Color Change and the ratings shall be based on the portion of the Black print exhibiting the most color loss. The lower of the two (2) ratings of each bag shall be recorded as the result for the bag. Failure of any of the five (5) bags to meet the required rating shall be considered a test failure.

4.4.11.3 Light. AATCC-16, Option A (after 40 fading units) or E (after 170 kilojoules).

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4.4.11.4 Abrasion (Black 357 only). AATCC-119 except that the number of abrasion cycles shall be 300.

4.4.12 Synthetic perspiration test. The specimen, 8-inches by 8-inches, shall be cut and exposed to synthetic perspiration as follows: The synthetic perspiration solution shall be made by combining 3.0 grams sodium chloride, 1.0 gram trypticase soy broth powder, 1.0 gram normal propyl propionate, 0.5 gram of liquid lecithin and 500 ml of distilled water. Cover the solution and stir while heating to 50°C until all ingredients are dissolved. Then, cool the solution to 35°C, remove cover and dispense it immediately with a pipette or other suitable measuring device. Dispense 2 ml of perspiration solution at 35°C onto the center of an 8-inch by 8-inch by ¼-inch glass plate. Place the specimen on the glass plate with the back side contacting the glass. Dispense an additional 2 ml of synthetic perspiration solution onto the center of the specimen. Place a second 8-inch by 8-inch by ¼-inch glass plate on top of the specimen and then place a 4-pound weight on top of and in the center of the assembly. After 16 hours, remove the specimen (do not rinse) and air dry the specimen before testing.

4.4.13 Contamination procedure. The specimen (or specimen area) shall be laid flat, face side up, on a glass plate. Three (3) drops of the test liquid shall be applied to the center of the specimen (or specimen area); as applicable, the test liquid shall be diethyltoluamide (see 6.6), diesel fuel (ASTM D-975, Grade 1-D), weapons lubricant (MIL-L-63460, see 6.7), motor oil (ASTM D-4485, Grade CD-II), and JP-8 fuel (MIL-T-83133). A glass plate of the same dimensions shall be placed on the specimen (or specimen area) and a pressure of 0.0625 pounds per square inch of glass plate contact area be applied to the assembly. After 16 hours, remove the specimen (or specimen area) from the assembly and test immediately for the required performance property in the center of specimen (or specimen area). For testing the “Initial” condition, the test specimen shall be as specified by the applicable test method. For testing the “After Laundering” condition, the laundering test shall be conducted in accordance with 4.4.14 for one laundering and drying cycle and one laundering sample, 48 inches by the full width of the cloth; the specimen areas for each test liquid shall be marked on the face side (using a laundry marker pen) for the specimen areas for hydrostatic resistance and for leakage; after laundering, the specimen areas may be cut for the laundering sample to facilitate performance property testing. Five (5) specimens (minimum) shall be tested for each of the “Initial” and the “After Laundering” conditions.

4.4.14 Laundering procedure. Place 2.0 ± 0.2 pounds of the cloth and, if needed, ballast in an automatic washing machine set on permanent press cycle, high water level and warm (100° + 10°F, -0°F) wash temperature. Place 0.5 ounce (14 grams) of 1993 AATCC Standard Reference Detergent (non-phosphate) without optical brighteners into the washer. The duration of each laundering cycle shall be 30 ± 5 minutes. After laundering, place sample and ballast in an automatic tumble dryer set on permanent press cycle, 150°-160°F, and dry for approximately fifteen (15) minutes. The laundering equipment, washer and dryer, shall be in accordance with AATCC-135.

4.4.15 Pattern execution. The pattern of the cloth shall be matched to the pattern drawing (see 6.3).

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4.4.16 Spectral reflectance. Spectral reflectance data shall be determined on the face side and shall be obtained from 600 to 860 nanometers (nm) for Class 1 and 4 cloths and 700 to 860 nanometers for Class 2 cloth, at 20 nm intervals on a spectrophotometer (see 6.4) relative to the barium sulfate standard, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g., magnesium oxide or vitolite tiles (see 6.5). The spectral band width shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CIE source A or CIE source D65. The specimen shall be measured as a single layer, backed with six layers of the same fabric and shade. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6 inches away from the selvage. The specimen shall be viewed at an angle no greater than 10 degrees from the normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified shall be considered a test failure.

4.4.16.1 Accelerated laundering (Spectral Reflectance Durability Test). Each color, except Black 357, of the camouflage pattern cloth shall be laundered separately in accordance with AATCC-61 (Option 3A) except that a 4-gram sample size shall be used (Note: A sample size large enough to evaluate the spectral reflectance shall be used) and that the procedure shall be conducted using (10) stainless steel spheres and the 1993 AATCC Standard Reference Detergent without optical brightener. The samples shall then be evaluated for spectral reflectance in accordance with 4.4.16.

4.4.17 Color Matching. The cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of $7500^{o} \pm 200^{o}K$, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at $2300^{o} \pm 200^{o}K$.

4.4.18 Toxicity assessment. The contractor must furnish information which certifies that the finished product is composed of materials which have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals (see 2.2.2) and a repeated insult human patch test (Modified Draize Procedure)(see 2.3). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free living individuals.

5. **PACKAGING**

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory

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Control Point's Department's System Command. Packaging data retrieval is available for the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The Type I cloth is intended for use in fabricating bivy covers for the Modular Sleeping Bag System (MSBS). The Type II cloth is used in the fabrication of the Wet Weather, Parka and Trousers, Rainsuit. The Type III cloth is intended for use in the fabrication of parkas and trousers for the Extended Cold Weather Clothing System – Second Generation (2GECWCS). The Type IV cloth is used in the fabrication of the parkas and trousers for the Extended Cold Weather Clothing System (ECWCS) and of the Air Force Cold Weather Security Police Jacket and Hood. The Type V cloth is intended for use in the fabrication of parkas and trousers for the Army's Extended Cold Weather Clothing System – Second Generation (2GECWCS-Army) Parka and Trousers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual document referenced (see 2.2.1 and 2.3).
c. Type and Class required.
d. When first article inspection is required (see 3.1), the item will be tested and should be a first article sample. The contacting officer should include specific instructions in acquisition documents regarding arrangement for examinations, quantity, and testing and approval.
e. Drawings required if applicable.

6.3 Standard sample. For access to samples and the pattern drawings, address the contracting activity issuing the invitation for bids.

6.4 Spectrophotometer. Suitable spectrophotometers for measuring spectral reflectance in the visible/near spectral include the Diano Hardy, Diano Match Scan, Hunter D54P-IR, and the MacBeth 1500 with IR options.

6.5 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available magnesium reagent (ribbon). Suitable tiles can be obtained for the National Institute of Standards and Technology or the instrument manufacturers.

6.6 Diethyltoluamide (DEET Insect Repellent) reagent. The insect repellent reagent is a solution of 75% by weight (min) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution is a technical grade and contains N, N-diethyl-metatoluamide of not less than 95% purity and the remainder shall consist of entirely or mixture

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of ortho or para isomers of N, N-diethyltoluamide. The denatured alcohol component of the solution is ethanol, U.S.P. 94.9% by volume and denatured in accordance with The Code of Federal Regulations 27 CFR 21, Formula 40 (see 2.1). The insect repellent must be registered with the U.S. Environment Protection Agency in accordance with the Federal Insecticide Act, Fungicide and Rodenticide (FIFRA)(see 2.1).

(For guidance purposes only, DEET insect repellent conforming to Type II, Concentration A of O-I-503 has been used successfully as a reagent in testing.)

6.7 Weapons lubricant MIL-L-63460 weapons lubricant is marketed as “Break-Free CLP” and may be obtained from Break-Free, Inc., Santa Anna, CA.

6.8 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

Cloth
Physical surface appearance
Rainsuit system
Bivy cover
Modular sleeping bag system (MSBS)
ECWCS (Extended Cold Weather Clothing System)
2GECWCS (Second Generation Extended Cold Weather Clothing System)
Jacket, Security Police

Custodian:
Army – GL
Navy – NU
Air Force – 99

Preparing activity:
DLA – CT

Project No. 8305-0856

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